

## Combinatorics Exam - January 2018

1. Let  $D(n)$  be the number of derangements of length  $n$ . Let  $F(n)$  be the number of permutations of length  $n$  that have exactly one fixed point. Determine (with proof) the number  $D(n) - F(n)$ .

2. Prove: If  $n$  and  $m$  are the number of vertices and edges in a simple planar graph, then  $m \leq 3n - 6$ .

Prove: For a simple graph  $G$  with  $n \geq 11$ , at most one of  $G$  or its complement  $\overline{G}$  is planar. (Recall that  $\overline{G}$  has the same vertex set as  $G$  and, for all vertices  $u, v$ , we have that  $\{u, v\}$  is an edge of  $\overline{G}$  if and only if  $\{u, v\}$  is not an edge of  $G$ .)

3. How many ways are there to seat  $n$  married couples at a straight table (only one side of the table) so that no woman sits next to her husband?

How many ways are there to seat  $n$  married couples at a straight table so that men and women alternate and no woman sits next to her husband?

4. Let  $B$  be a rectangular 3-dimensional box for which the width is equal to the length, but the depth is not equal to the length. How many ways are there to color the vertices of  $B$  using colors red and blue, where two colorings are considered the same if one can be obtained from the other by an orientation preserving symmetry of  $B$ .

5. Consider a directed graph obtained by putting a direction on each edge of the complete graph. Such a digraph is called a *tournament*. Prove that a tournament has a Hamiltonian path. Is it true that every tournament has a Hamiltonian cycle? Explain?

6. Let  $n \geq 3$ . Let  $h_n$  be the number of permutations of length  $n$  in which the number of inversions is divisible by three. Find an explicit formula for  $h_n$ .

7. Prove that the language  $\mathcal{L} = \{ww : w \in \{\mathbf{a}, \mathbf{b}\}^*\}$  is not regular.

8. Suppose  $\mathcal{C}$  and  $\mathcal{D}$  are permutation classes. Prove that  $\mathcal{C} \cup \mathcal{D}$  is a permutation class, and describe (with an algorithm, perhaps) how to compute its basis.